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1.Case Study: Monthly Hospital Patient Visits

A hospital administrator wants to analyze the number of patients visiting the hospital each month during 2025.

Month	Patient
Jan	1200
Feb	1350
Mar	1500
Apr	1450
May	1700
Jun	1850

Questions

1. Which visualization is most suitable?
Answer: Line Chart (or Line Graph)

2. What trend can be observed?
Answer: The number of patient visits shows an overall increasing trend from January to June, with a slight decrease in April.

3. Identify the month with the highest patient count.
Answer: June – 1850 patients

4. Predict whether visits are increasing or decreasing.
Answer: The visits are increasing overall and are likely to continue increasing.

2.Case Study: Study Hours vs Exam Marks

A teacher wants to determine whether studying more hours improves exam performance.

Student	Study Hours	Marks
A	2	45
B	3	52
C	4	58

D	5	68
E	7	82

Questions

1. Which chart is appropriate?
Answer: Scatter Plot

2. Is there a positive or negative relationship?
Answer: There is a positive relationship, as marks increase with study hours.

3. Which student is an outlier, if any?
Answer: There is no significant outlier in the data.

4. Can marks be predicted from study hours?
Answer: Yes. Since there is a positive correlation, marks can be predicted using study hours.

3.Case Study: Employee Salary Distribution

A company wants to understand how salaries are distributed among employees.

Salary Data (₹ in Thousands)

25, 28, 30, 32, 35, 35, 36, 38, 40, 42, 45, 48, 50, 52, 55, 60, 65, 70

Questions

1. Which graph should be used?
Answer: Histogram

2. In which salary range do most employees fall?
Answer: Most employees fall in the ₹30,000–₹50,000 salary range.

3. Is the distribution symmetric or skewed?
Answer: The distribution is slightly positively (right) skewed.

4. Are there any salary ranges with very few employees?
Answer: Yes. The ₹60,000–₹70,000 range has fewer employees.

4.Case Study: Favorite Programming Language

A survey was conducted among 200 students.

Language	**Number of Students**
Python	80
Java	50
C++	30

JavaScript	25
R	15

Questions

1. Which visualization is suitable?
Answer: Bar Chart (or Pie Chart)

2. Which language is most popular?
Answer: Python (80 students)

3. Which language is least preferred?
Answer: R (15 students)

4. Compare Python and Java popularity.
Answer: Python is more popular than Java by 30 students (80 vs 50).

5.Case Study: Household Monthly Expenses

A family spends money in the following categories:

Category	**Expense (%)**
Rent	40
Food	25
Education	15
Transportation	10
Entertainment	10

Questions

1. Which chart best represents the data?
Answer: Pie Chart

2. Which category consumes the largest budget?
Answer: Rent (40%)

3. What percentage is spent on Food and Education together?
Answer: 40% (Food 25% + Education 15%)

4. Which categories occupy the smallest share?
Answer: Transportation and Entertainment (10% each)

6.Case Study: Student Test Scores

Scores obtained by students:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Questions

1. Which plot should be used?
Answer: Box Plot
2. What is the median score?
Answer: 65
3. Are there any outliers?
Answer: Yes. 95 is a potential outlier.
4. What is the spread of the data?
Answer: The scores range from 45 to 95, giving a range of 50 marks, which indicates a moderate spread.